

LAB EXPERIMENTS

TOOL:R Programming

RegisterNo:192424335

Name: M.Rishitha

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COURSE:Data Handling and Visualization

Code: DSA0604

SET-6

Question 1: Grouped Bar Chart

R PROGRAM:

```
library(ggplot2)
library(tidyr)
sales <- data.frame(
  Product = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)
sales_long <- pivot_longer(
  sales,
  cols = January:March,
  names_to = "Month",
  values_to = "Sales"
)
ggplot(sales_long,
  aes(x = Product,
    y = Sales,
    fill = Month)) +
  geom_bar(stat = "identity",
    position = "dodge") +
  labs(
    title = "Monthly Product Sales (First Quarter)",
    x = "Product",
```

```

y = "Sales"
) +
theme_minimal()

```

OUTPUT:



Question 2: Stacked Area Chart

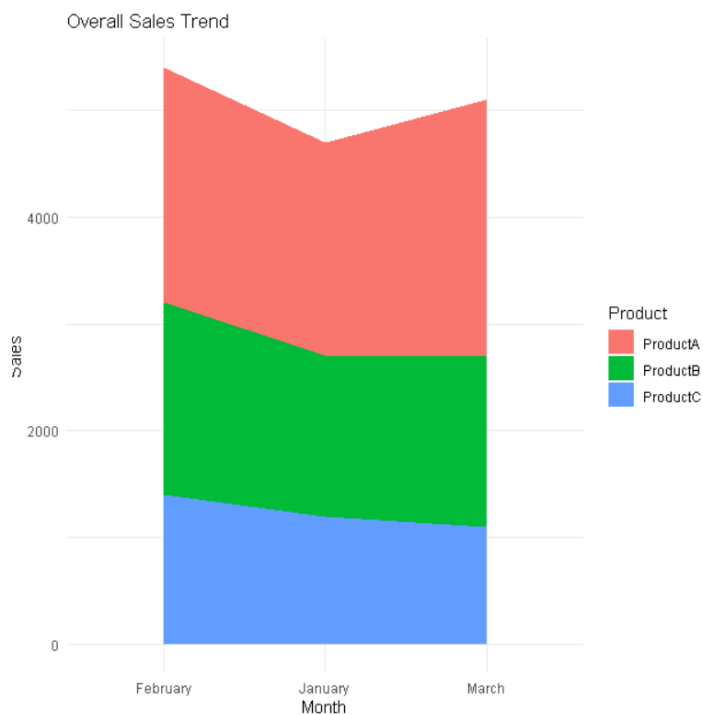
R PROGRAM:

```

library(ggplot2)
library(tidyr)
sales <- data.frame(
  Month = c("January", "February", "March"),
  ProductA = c(2000, 2200, 2400),
  ProductB = c(1500, 1800, 1600),
  ProductC = c(1200, 1400, 1100)
)
sales_long <- pivot_longer(
  sales,
  cols = ProductA:ProductC,
  names_to = "Product",
  values_to = "Sales"
)

```

```
)
ggplot(sales_long,
  aes(x = Month,
      y = Sales,
      fill = Product,
      group = Product)) +
  geom_area() +
  labs(
    title = "Overall Sales Trend",
    x = "Month",
    y = "Sales"
  ) +
  theme_minimal()
```



Question 3: Monthly Sales Table

R PROGRAM:

```
sales <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),
```

```

JanuarySales = c(2000,1500,1200),
FebruarySales = c(2200,1800,1400),
MarchSales = c(2400,1600,1100)
)
print(sales)

print(sales)
ProductID ProductName JanuarySales FebruarySales MarchSales
1 Product A 2000 2200 2400
2 Product B 1500 1800 1600
3 Product C 1200 1400 1100
# Question 3: Monthly Sales Table

sales <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),
  JanuarySales = c(2000,1500,1200),
  FebruarySales = c(2200,1800,1400),
  MarchSales = c(2400,1600,1100)
)

print(sales)
ProductID ProductName JanuarySales FebruarySales MarchSales
1 Product A 2000 2200 2400
2 Product B 1500 1800 1600
3 Product C 1200 1400 1100

```

Question 4: Interactive Dashboard (Shiny + Plotly)

R PROGRAM:

```
library(shiny)
```

```
library(plotly)
```

```
library(tidyr)
```

```

sales <- data.frame(
  Product = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)

```

```
sales_long <- pivot_longer(
```

```
    sales,
    cols = January:March,
    names_to = "Month",
    values_to = "Sales"
  )

  ui <- fluidPage(

    titlePanel("Product Sales Dashboard"),

    fluidRow(

      column(
        6,
        plotlyOutput("barPlot")
      ),

      column(
        6,
        plotlyOutput("areaPlot")
      )
    ),

    br(),

    tableOutput("salesTable")
  )
```

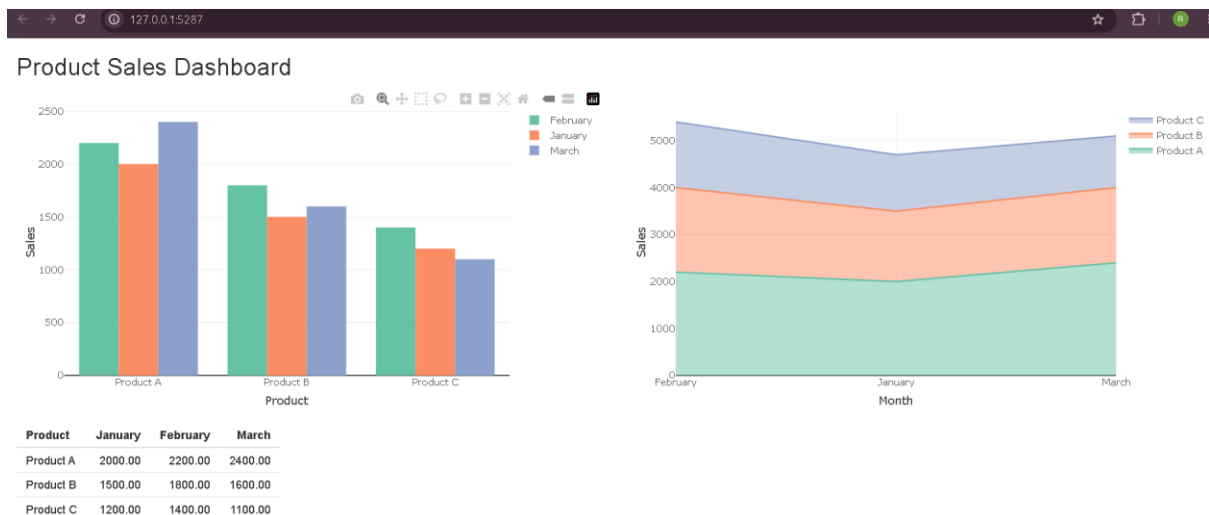
```

server <- function(input, output){
  output$barPlot <- renderPlotly({
    plot_ly(
      sales_long,
      x = ~Product,
      y = ~Sales,
      color = ~Month,
      type = "bar"
    ) %>%
      layout(barmode = "group")

  }) output$areaPlot <- renderPlotly({
    plot_ly(
      sales_long,
      x = ~Month,
      y = ~Sales,
      color = ~Product,
      type = "scatter",
      mode = "lines",
      stackgroup = "one"
    )
  })
  output$salesTable <- renderTable({
    sales
  })
}

shinyApp(ui, server)

```



SET-7

Question 1: Bar Chart - Customer Age Distribution

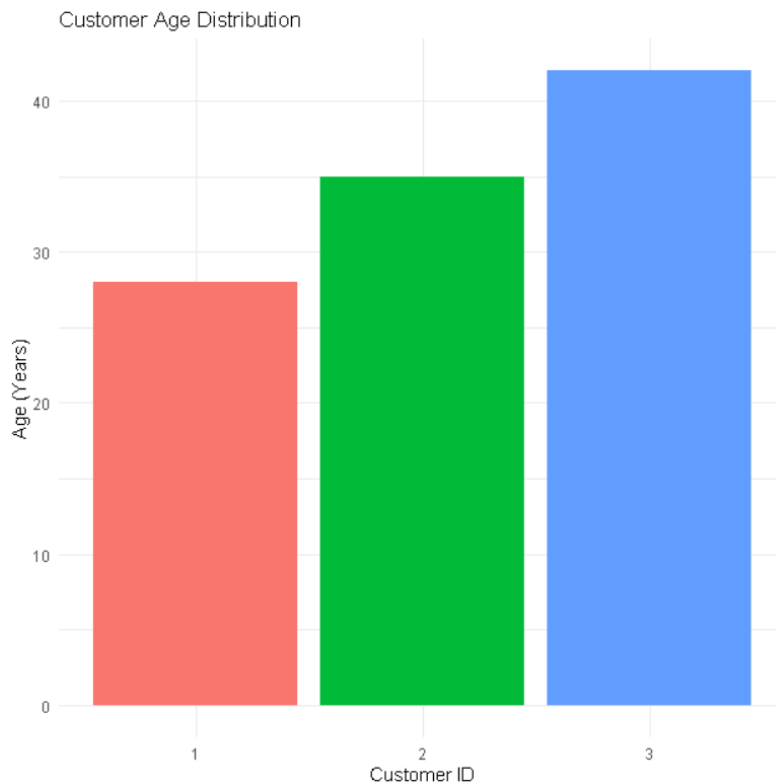
R PROGRAM:

```
library(ggplot2)

customers <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

ggplot(customers,
  aes(x = factor(CustomerID),
    y = Age,
    fill = factor(CustomerID))) +
  geom_bar(stat = "identity") +
  labs(
    title = "Customer Age Distribution",
    x = "Customer ID",
    y = "Age (Years)"
  ) +
```

```
theme_minimal() +
theme(legend.position = "none")
```



Question 2: Pie Chart - Gender Distribution

R PROGRAM

```
library(ggplot2)
customers <- data.frame(
  Gender = c("Female", "Male", "Female")
)
gender_count <- as.data.frame(table(customers$Gender))
colnames(gender_count) <- c("Gender", "Count")
ggplot(gender_count,
  aes(x = "",
    y = Count,
    fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(
```

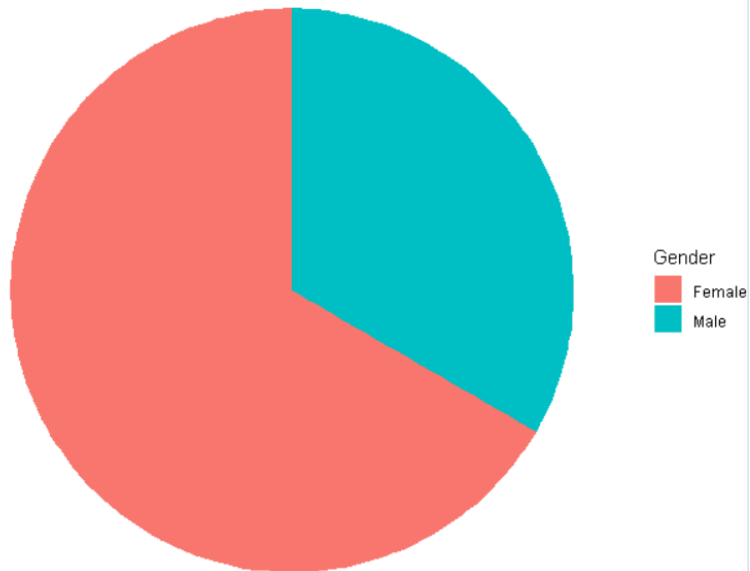


```

    title = "Customer Gender Distribution"
  ) +
  theme_void()

```

Customer Gender Distribution



Question 3: Customer Demographics Table

R PROGRAM:

```

customers <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
print(customers)

```

```

+ )
>
> print(customers)
  CustomerID Age Gender Income
1          1  28 Female  50000
2          2  35  Male  60000
3          3  42 Female  75000
> |

```

Question 4: Interactive Dashboard (Shiny + Plotly)

R PROGRAM:

```

library(shiny)
library(plotly)

customers <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

gender_count <- as.data.frame(table(customers$Gender))
colnames(gender_count) <- c("Gender","Count")

\ui <- fluidPage(
  titlePanel("Customer Demographics Dashboard"),
  fluidRow(
    column(
      6,
      plotlyOutput("barPlot")
    ),
    column(
      6,
      plotlyOutput("piePlot")
    )
  )
)

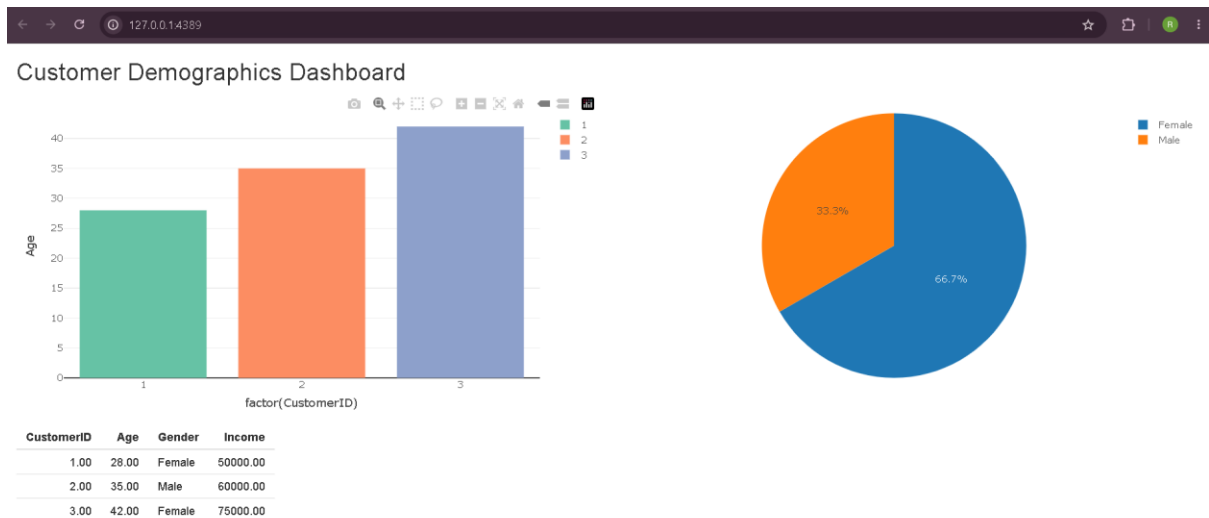
```

),

br(),

tableOutput("customerTable")

```
)server <- function(input, output){  
  output$barPlot <- renderPlotly({  
    plot_ly(  
      customers,  
      x = ~factor(CustomerID),  
      y = ~Age,  
      type = "bar",  
      color = ~factor(CustomerID)  
    )  
  })output$piePlot <- renderPlotly({  
    plot_ly(  
      gender_count,  
      labels = ~Gender,  
      values = ~Count,  
      type = "pie"  
    )  
  })  
  
  output$customerTable <- renderTable({  
    customers  
  })  
}  
shinyApp(ui, server)
```



SET-8

Question 1: Line Chart - Employee Performance Trend

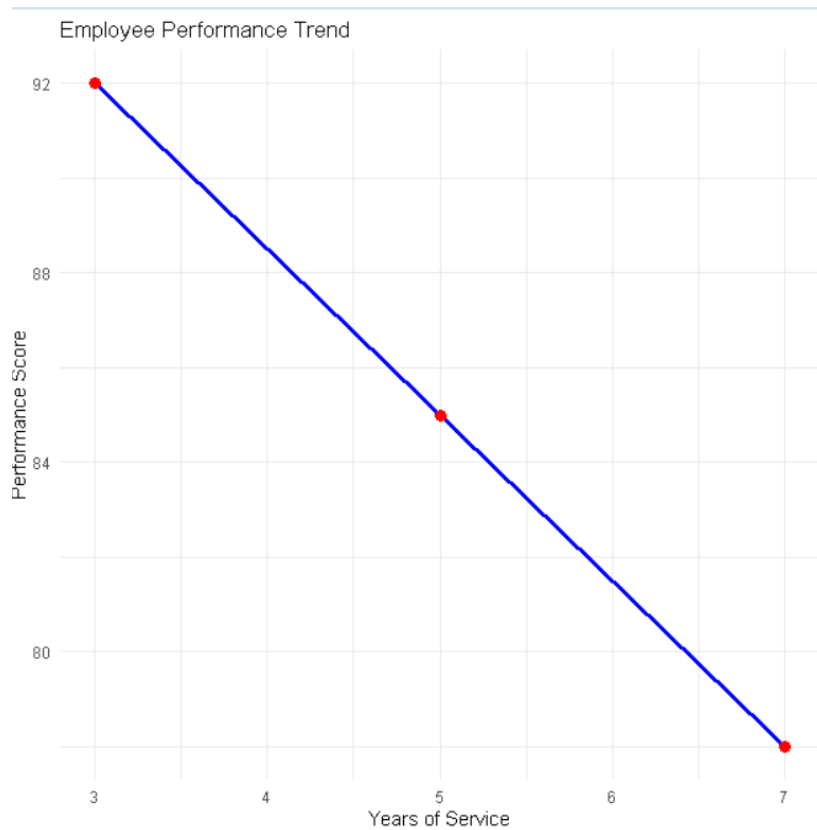
R PROGRAM:

```
library(ggplot2)

employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsOfService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)

ggplot(employee,
  aes(x = YearsOfService,
      y = PerformanceScore,
      group = 1)) +
  geom_line(color = "blue", linewidth = 1.2) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Employee Performance Trend",
    x = "Years of Service",
    y = "Performance Score"
```

) +

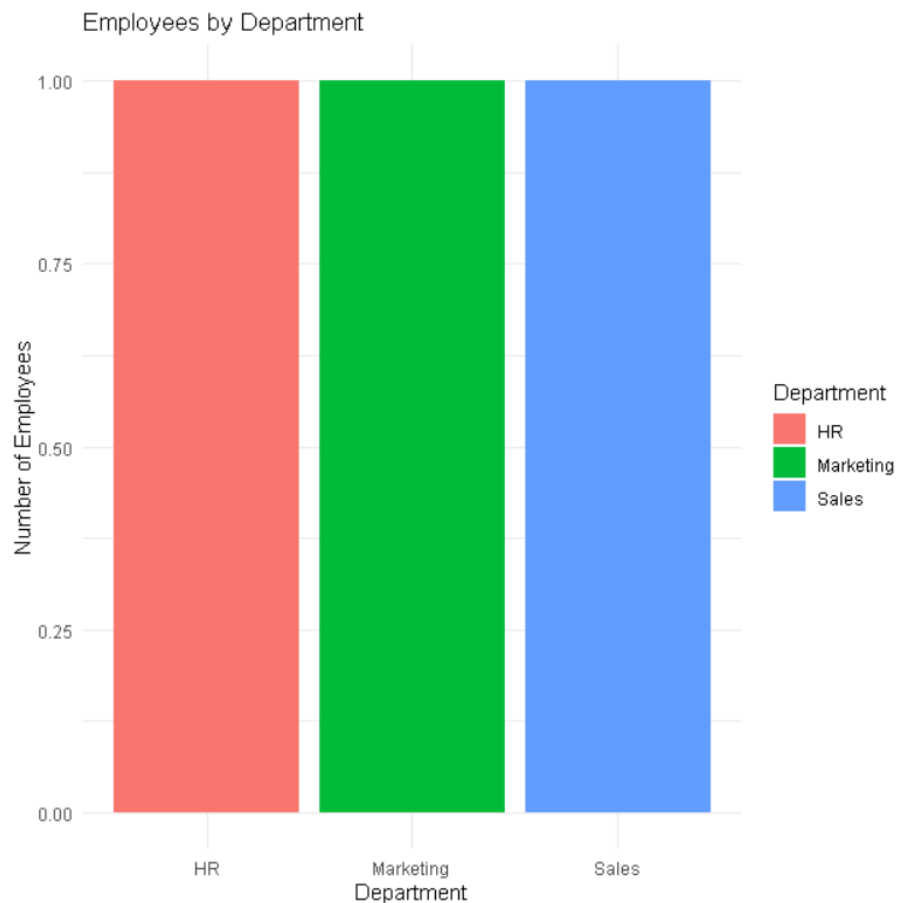


Question 2: Bar Chart - Employees by Department

R PROGRAM:

```
library(ggplot2)
employee <- data.frame(
  Department = c("Sales","HR","Marketing")
)
dept_count <- as.data.frame(table(employee$Department))
colnames(dept_count) <- c("Department","Count")
ggplot(dept_count,
  aes(x = Department,
    y = Count,
    fill = Department)) +
  geom_bar(stat = "identity") +
```

```
labs(
  title = "Employees by Department",
  x = "Department",
  y = "Number of Employees"
) +
theme_minimal()
```



Question 3: Employee Performance Table

R PROGRAM:

```
employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsOfService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
print(employee)
```

```

> employee <- data.frame(
+   EmployeeID = c(1,2,3),
+   Department = c("Sales","HR","Marketing"),
+   YearsOfService = c(5,3,7),
+   PerformanceScore = c(85,92,78)
+ )
>
> print(employee)
  EmployeeID Department YearsOfService PerformanceScore
1          1     Sales              5                85
2          2        HR              3                92
3          3  Marketing              7                78

```

Question 4: Interactive Dashboard (Shiny + Plotly)

R PROGRAM:

```

library(shiny)
library(plotly)

employee <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsOfService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)

dept_count <- as.data.frame(table(employee$Department))
colnames(dept_count) <- c("Department","Count")

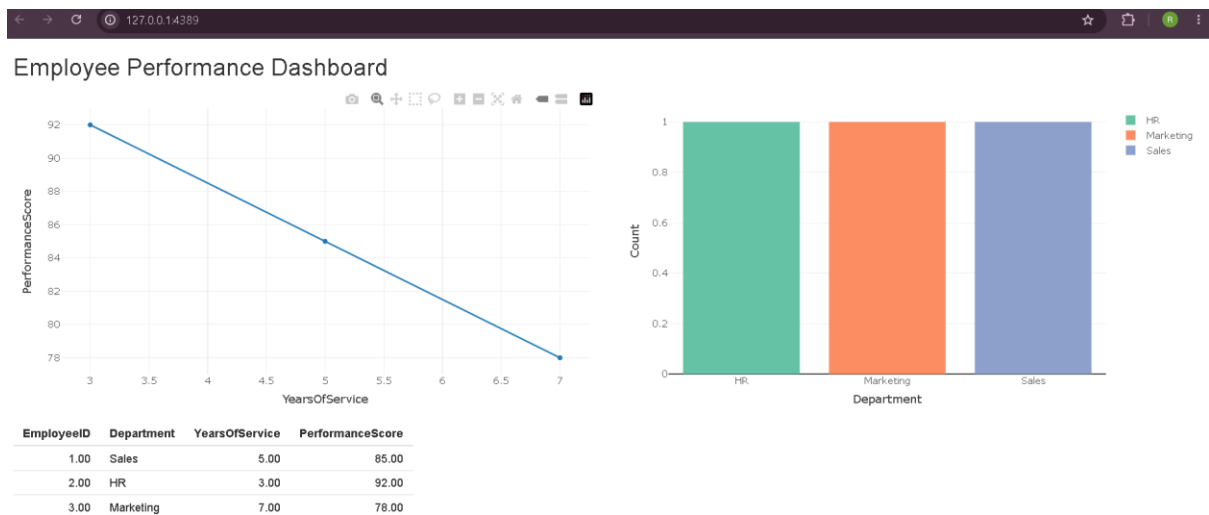
ui <- fluidPage(
  titlePanel("Employee Performance Dashboard"),
  fluidRow(
    column(
      6,
      plotlyOutput("linePlot")
    ),
    column(
      6,
      plotlyOutput("barPlot")
    )
  )
)

```

```

),
br(),
tableOutput("employeeTable"))
server <- function(input, output){
  output$linePlot <- renderPlotly({
    plot_ly(
      employee,
      x = ~YearsOfService,
      y = ~PerformanceScore,
      type = "scatter",
      mode = "lines+markers"
    )
  })
  output$barPlot <- renderPlotly({
    plot_ly(
      dept_count,
      x = ~Department,
      y = ~Count,
      type = "bar",
      color = ~Department
    )
  })
  output$employeeTable <- renderTable({
    employee
  })
}
shinyApp(ui, server)

```

SET -9

Question 1: Import Dataset, Histogram & Boxplot

R PROGRAM:

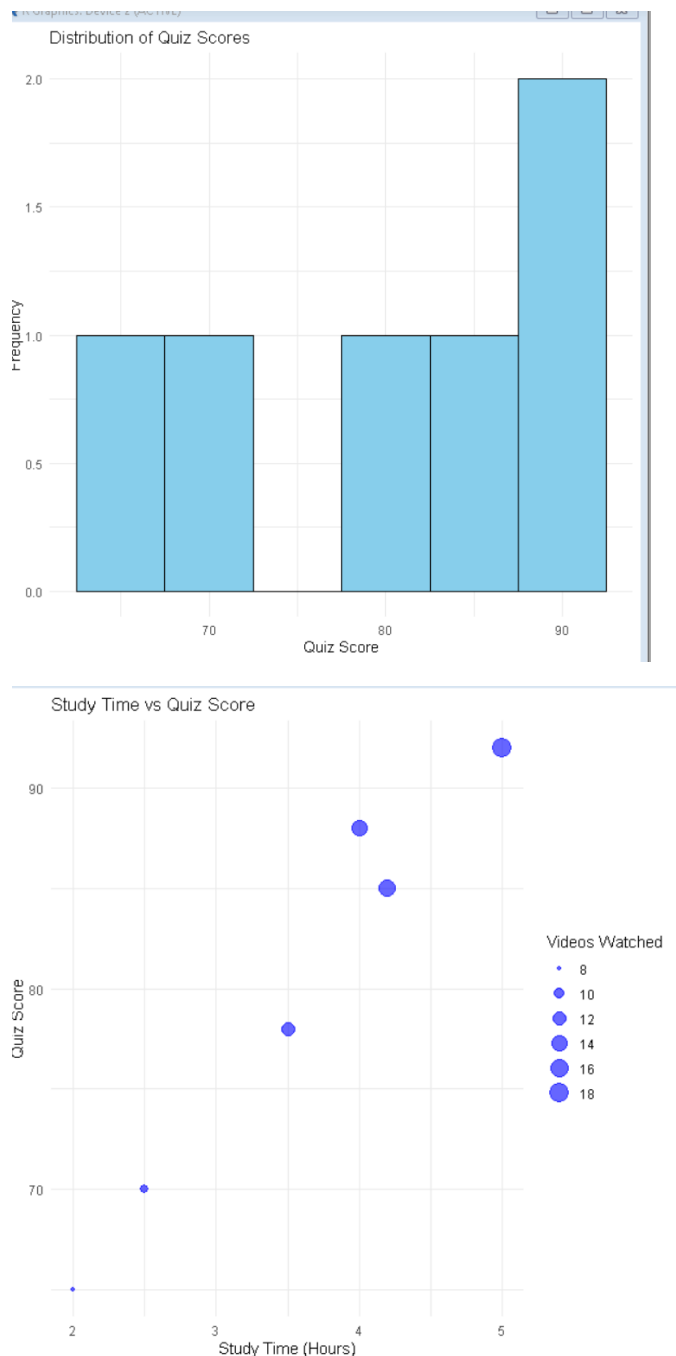
```
library(ggplot2)

data <- read.csv("Online_Learning_Activity.csv")

ggplot(data, aes(x = Quiz_Score)) +
  geom_histogram(binwidth = 5,
                 fill = "skyblue",
                 color = "black") +
  labs(
    title = "Distribution of Quiz Scores",
    x = "Quiz Score",
    y = "Frequency"
  ) +
  theme_minimal()

ggplot(data,
  aes(x = Course,
      y = Quiz_Score,
      fill = Course)) +
  geom_boxplot() +
```

```
labs(
  title = "Quiz Score by Course",
  x = "Course",
  y = "Quiz Score"
) +
theme_minimal()
```



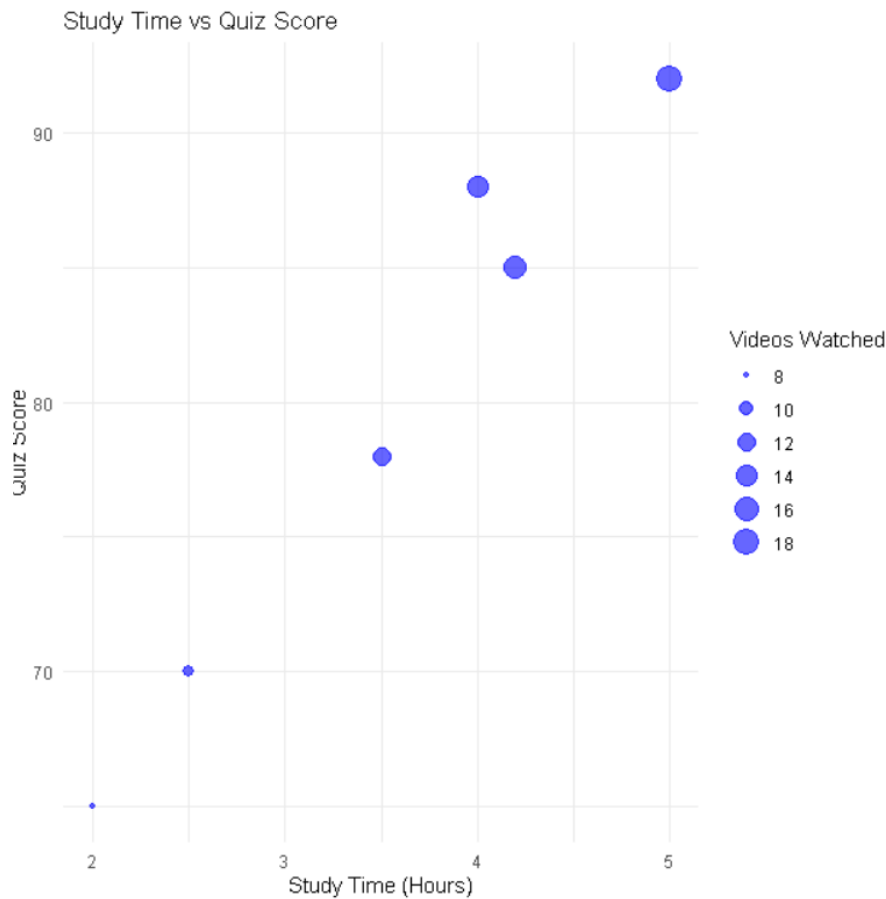
Question 2: Bubble Scatter Plot

R PROGRAM:

```
library(ggplot2)

data <- read.csv("Online_Learning_Activity.csv")

ggplot(data,
  aes(x = Study_Time,
      y = Quiz_Score,
      size = Videos_Watched)) +
  geom_point(alpha = 0.6,
    color = "blue") +
  labs(
    title = "Study Time vs Quiz Score",
    x = "Study Time (Hours)",
    y = "Quiz Score",
    size = "Videos Watched"
  ) +
  theme_minimal()
```



Question 3: Monthly Average Quiz Score with Moving Average

R PROGRAM:

```
library(ggplot2)
library(dplyr)
library(zoo)

data <- read.csv("Online_Learning_Activity.csv")
data$Login_Date <- as.Date(data$Login_Date)
data$Month <- format(data$Login_Date, "%Y-%m")

monthly <- data %>%
  group_by(Month) %>%
  summarise(Average_Quiz = mean(Quiz_Score))

monthly$Moving_Average <- rollmean(
  monthly$Average_Quiz,
  k = 2,
  fill = NA,
```

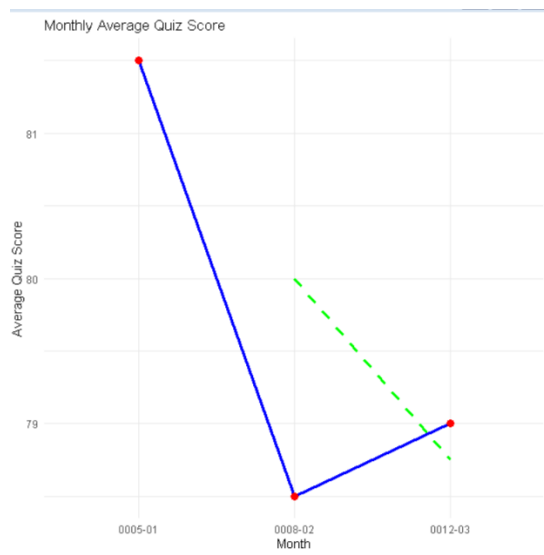
```
align = "right"
)
ggplot(monthly,
  aes(x = Month,
      y = Average_Quiz,
      group = 1)) +
geom_line(color = "blue",
  linewidth = 1.2) +

geom_point(size = 3,
  color = "red") +

geom_line(aes(y = Moving_Average),
  color = "green",
  linewidth = 1.2,
  linetype = "dashed") +

labs(
  title = "Monthly Average Quiz Score",
  x = "Month",
  y = "Average Quiz Score"
) +

theme_minimal()
```



Question 4: Interactive Dashboard (Shiny + Plotly)

R PROGRAM:

```
library(shiny)
library(plotly)
data <- read.csv("Online_Learning_Activity.csv")
ui <- fluidPage(
  titlePanel("Online Learning Dashboard"),
  sidebarLayout(
    sidebarPanel(
      selectInput(
        "gender",
        "Select Gender",
        choices = c("All",
                     unique(data$Gender))
      ),
      selectInput(
        "course",
        "Select Course",
        choices = c("All",
```

```

        unique(data$Course))
    )

    ),
  mainPanel(
    plotlyOutput("barPlot"),
    br(),plotlyOutput("scatterPlot")

  )

)

)

)

server <- function(input, output){

  filtered <- reactive({

    df <- data

    if(input$gender != "All")
      df <- df[df$Gender == input$gender,]

    if(input$course != "All")
      df <- df[df$Course == input$course,]

    df

  })

```

```

output$barPlot <- renderPlotly({

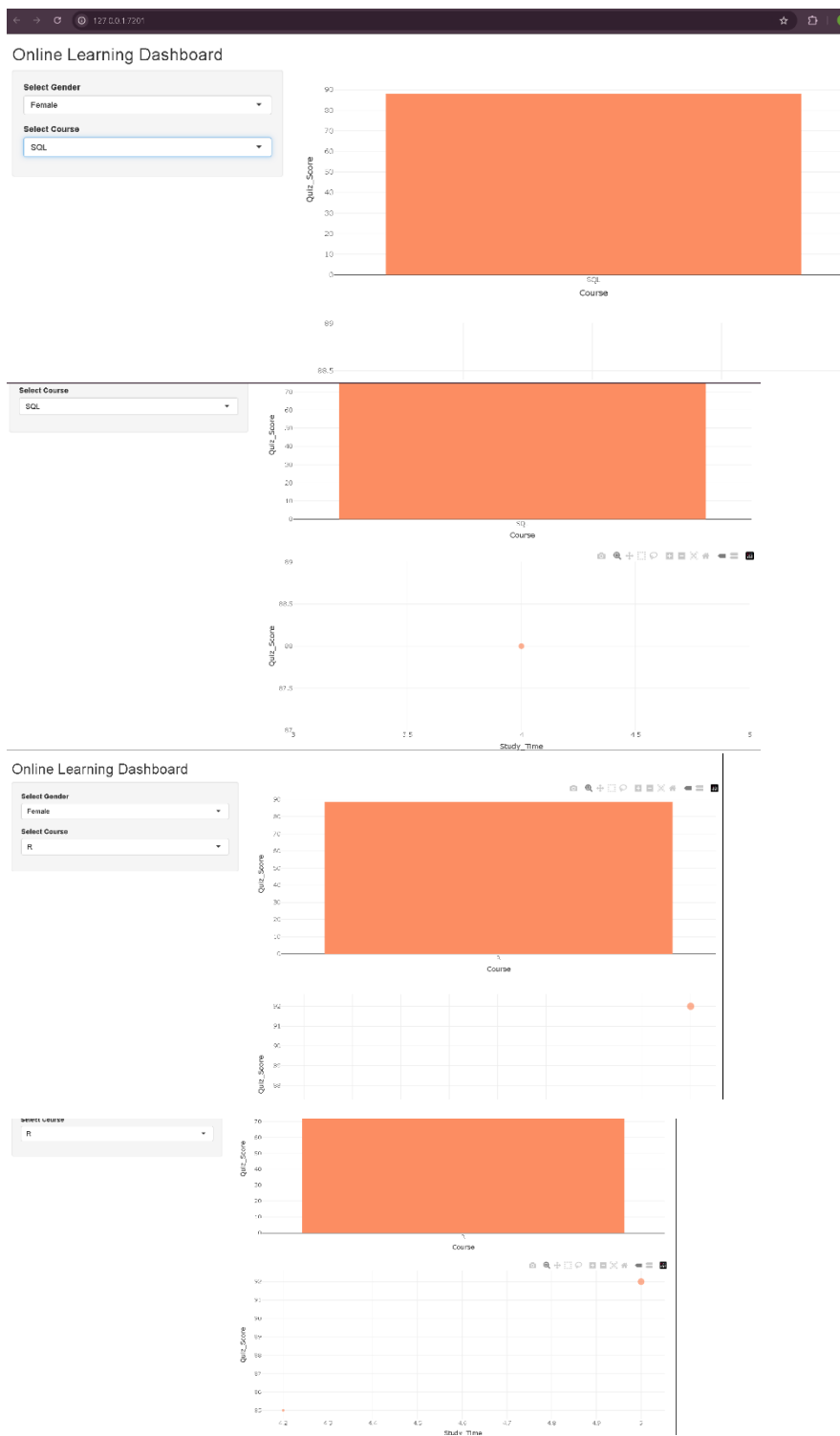
  avg <- aggregate(
    Quiz_Score ~ Course,
    data = filtered(),
    mean
  )

  plot_ly(
    avg,
    x = ~Course,
    y = ~Quiz_Score,
    type = "bar",
    color = ~Course
  )

})output$scatterPlot <- renderPlotly({ plot_ly(
  filtered(),
  x = ~Study_Time,
  y = ~Quiz_Score,
  type = "scatter",
  mode = "markers",
  size = ~Videos_Watched,
  color = ~Course
)
})
}

shinyApp(ui, server)

```

SET -10

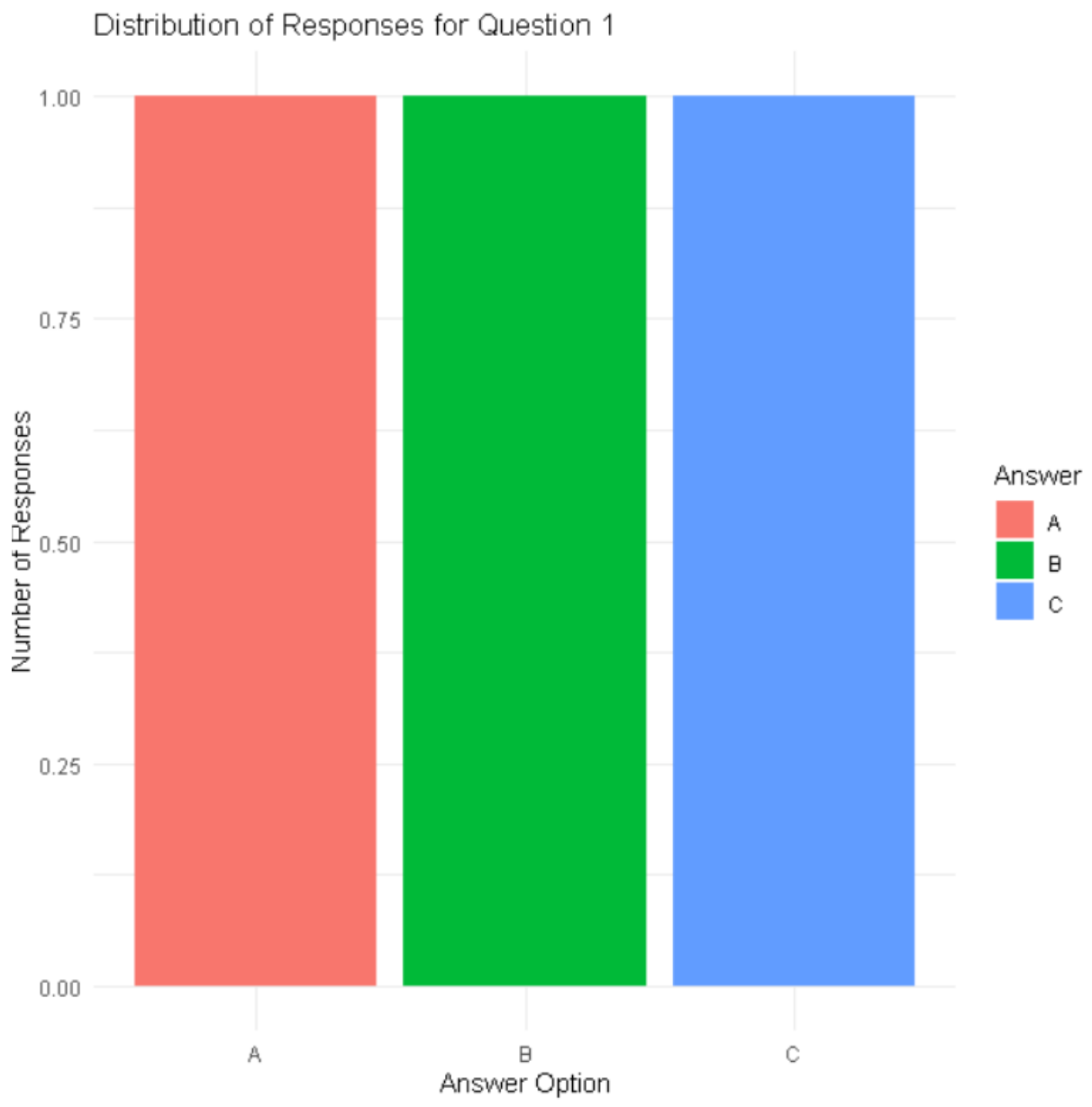
Question 1: Grouped Bar Chart - Question 1 Responses

```
library(ggplot2)
```

```
survey <- data.frame(
```

```
  SurveyID = c(1,2,3),
```

```
Question1 = c("A","B","C"),
Question2 = c("B","A","A"),
Question3 = c("C","D","B")
)
q1_count <- as.data.frame(table(survey$Question1))
colnames(q1_count) <- c("Answer","Count")
ggplot(q1_count,
  aes(x = Answer,
      y = Count,
      fill = Answer)) +
  geom_bar(stat = "identity",
    position = "dodge") +
  labs(
    title = "Distribution of Responses for Question 1",
    x = "Answer Option",
    y = "Number of Responses"
  ) +
  theme_minimal()
```



Question 2

```
library(ggplot2)
library(tidyr)
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
```

```

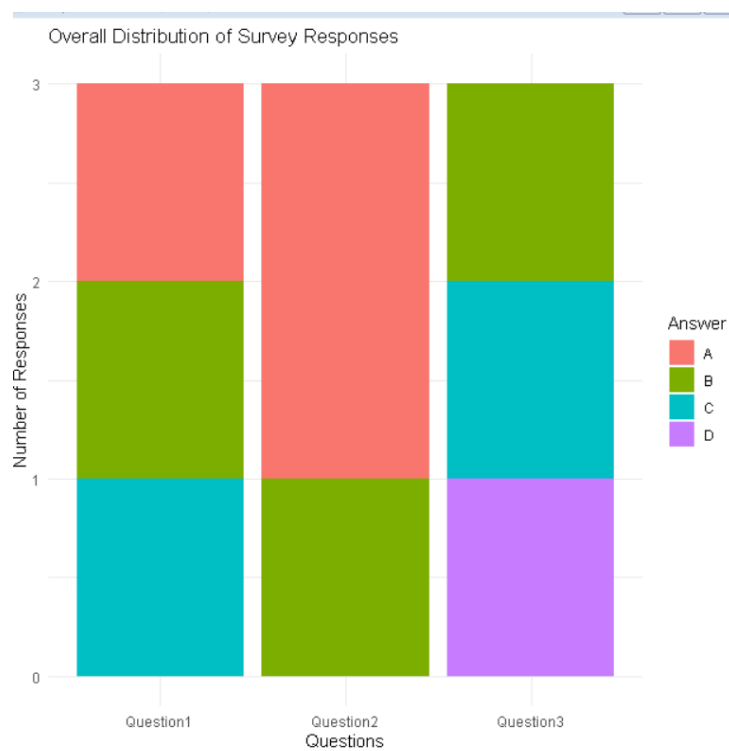
survey_long <- pivot_longer(
  survey,
  cols = Question1:Question3,
  names_to = "Question",
  values_to = "Answer"
)

```

```

ggplot(survey_long,
  aes(x = Question,
    fill = Answer)) +
  geom_bar() +
  labs(
    title = "Overall Distribution of Survey Responses",
    x = "Questions",
    y = "Number of Responses"
  ) +
  theme_minimal()

```



Question 3: Survey Response Table

R program:

```
survey <- data.frame(  
  SurveyID = c(1,2,3),  
  Question1 = c("A","B","C"),  
  Question2 = c("B","A","A"),  
  Question3 = c("C","D","B")  
)  
  
print(survey)  
+ )  
>  
> print(survey)  
  SurveyID Question1 Question2 Question3  
1         1         A         B         C  
2         2         B         A         D  
3         3         C         A         B  
> |
```

Question 4: Interactive Dashboard (Shiny + Plotly)

R PROGRAM:

```
library(shiny)  
library(plotly)  
library(tidyr)  
  
survey <- data.frame(  
  SurveyID = c(1,2,3),  
  Question1 = c("A","B","C"),  
  Question2 = c("B","A","A"),  
  Question3 = c("C","D","B")  
)  
  
q1_count <- as.data.frame(table(survey$Question1))  
colnames(q1_count) <- c("Answer","Count")  
  
survey_long <- pivot_longer(  
  survey,
```

```

cols = Question1:Question3,
names_to = "Question",
values_to = "Answer"
)
response_count <- as.data.frame(table(survey_long$Question,
                                     survey_long$Answer))
colnames(response_count) <- c("Question", "Answer", "Count")

```

```

ui <- fluidPage(
  titlePanel("Survey Responses Dashboard"),
  fluidRow(
    column(
      6,
      plotlyOutput("groupBar")
    ),
    column(
      6,
      plotlyOutput("stackBar")
    )
  ),
  br(),
  tableOutput("surveyTable")
)

```

```

server <- function(input, output){
  output$groupBar <- renderPlotly({
    plot_ly(
      q1_count,
      x = ~Answer,
      y = ~Count,
      type = "bar",

```

```

    color = ~Answer
  )
})
output$stackBar <- renderPlotly({
plot_ly(
  response_count,
  x = ~Question,
  y = ~Count,
  color = ~Answer,
  type = "bar"
) %>%
  layout(barmode = "stack")
})
output$surveyTable <- renderTable({
  survey
})
}

shinyApp(ui, server)

```

